Notes: Scharfstein & Stein (AER, 1990)
Herd Behavior and Investment

Contents

1 Big picture

- **Question.** Why do professional managers, investors, or decision makers sometimes ignore their own private information and imitate others, even when private information is substantively useful?
- **Core answer.** Herding can be individually rational when managers care about their reputations in a labor market that is trying to infer their ability from actions and outcomes. A manager may prefer to be wrong in the same way as others rather than risk being wrong alone.
- **Main mechanism.** Managers are uncertain about whether they are "smart" or "dumb." Smart managers receive informative signals, while dumb managers receive uninformative signals. If smart managers' errors are correlated, then deviating from earlier managers is risky for reputation because the market interprets disagreement as evidence of low ability.
- **Model structure.** Two firms, A and B, invest sequentially. Manager A moves first. Manager B observes A's action and receives his own private signal. The labor market observes actions and outcomes and then updates beliefs about managerial ability.
- **Central result.** There is no continuation equilibrium in which manager B's action is truly signal-contingent. Instead, B either mimics A or contradicts A regardless of B's own signal. With reasonable off-equilibrium beliefs, the natural equilibrium is herding: B mimics A.
- **Inefficiency.** Herding destroys information aggregation. B's private signal is socially valuable, but it is not revealed through B's action. Aggregate investment therefore fails to reflect all available information.
- **Applications.** The paper applies the model to corporate investment, bank lending to developing countries, stock-market investment by money managers, and internal decision making within firms.
- **Important qualification.** Herding is not inevitable. Profit-based incentives, limited liability, relative rather than absolute ability concerns, and alternative definitions of managerial talent can weaken herding incentives.
- **Economic takeaway.** Reputation can create conformity. The model gives a rational, career-concern-based foundation for Keynes's idea that it is often safer for professional investors to "fail conventionally" than to "succeed unconventionally."

2 Incremental contribution

- **Contribution to herd-behavior theory.** Earlier discussions of herding were often psychological or informal. Scharfstein and Stein provide a formal equilibrium model in which herding is fully rational for career-concerned managers.
- **Contribution to agency theory.** The paper shows that agency problems can distort information aggregation even without direct conflicts over cash flows. Managers may choose actions to influence beliefs about their ability rather than to maximize firm value.
- **Contribution relative to classical investment theory.** Classical models assume decisions reflect all available information. The paper shows that, with reputational concerns, available private information may be intentionally ignored.
- **Contribution relative to rational expectations models.** In noisy rational expectations settings, agents learn from others' actions or prices. Scharfstein and Stein show that the actions themselves may stop revealing information because reputational incentives induce pooling on the herd action.
- **Contribution to corporate finance.** The model gives a mechanism for correlated corporate investment mistakes: managers may imitate peers' capital-allocation decisions to share blame and protect labor-market reputation.

- **Contribution to asset-pricing/market microstructure.** The model gives microfoundations for trend-following and institutional imitation. Prices need not reflect dispersed information if professional money managers prefer conventional behavior.
- **Contribution to organizational economics.** The paper explains why voting order, committee design, and information flow within firms matter. If junior managers have stronger reputational concerns, bottom-up information aggregation may outperform top-down decision making.
- **Methodological contribution.** The paper demonstrates that correlated prediction errors among skilled agents are sufficient to break signal-contingent equilibria. Herding is not driven by stupidity or irrationality; it is driven by Bayesian updating plus career incentives.
- **Policy and governance contribution.** The paper suggests that incentive contracts, decentralization, anonymous/simultaneous reporting, and organizational design can mitigate herding by reducing the reputational penalty of dissent.

3 Model objects and measurement

3.1 Players, timing, and investment environment

- There are two firms, firm A and firm B, run by managers A and B.
- Investment occurs sequentially. Manager A moves at date 1; manager B observes A's action and moves at date 2; the state is realized at date 3.
- Each firm can invest in a project whose payoff is either high or low.
- The high state yields net profit $X_H > 0$; the low state yields net profit $X_L < 0$.
- The prior probability of the high state is α; the low-state probability is $1 - \alpha$.
- The investment is available at a fixed price, so the model abstracts from price feedback and market clearing.

Interpretation: The fixed-price assumption makes the model most directly applicable to real investment projects, such as adoption of a technology or lending to a country. In stock-market applications, prices would move with demand, but the same reputational logic can still operate through asset allocation decisions.

3.2 Managerial types: smart and dumb

- A manager is smart with prior probability θ.
- A manager is dumb with probability $1 - \theta$.
- Neither the manager nor the labor market initially knows the manager's type.
- A smart manager receives an informative signal about the state.
- A dumb manager receives a signal that is statistically uninformative about the state.

Interpretation: The labor market's problem is to infer ability from actions and outcomes. Managers care about this inference because it determines future wages or career opportunities.

3.3 Signals and likelihoods

Signals are either good, S_G, or bad, S_B. For smart managers,

$$\text{Prob}(S_G \mid X_H, \text{smart}) = p, \qquad \text{Prob}(S_G \mid X_L, \text{smart}) = q < p. \tag{1}$$

For dumb managers,

$$\text{Prob}(S_G \mid X_H, \text{dumb}) = \text{Prob}(S_G \mid X_L, \text{dumb}) = z. \tag{2}$$

The paper assumes the unconditional distribution of signals is the same for smart and dumb managers:

$$z = \alpha p + (1 - \alpha)q. \tag{3}$$

Interpretation: A manager's signal does not directly reveal whether he is smart. Only the match between actions, outcomes, and other managers' actions helps the labor market update beliefs about ability.

3.4 Posterior beliefs about project quality

After receiving a good signal, the manager's posterior probability of the high state is

$$\mu_G = \text{Prob}(X_H \mid S_G) = \frac{[\theta p + (1 - \theta)z]\alpha}{z}. \tag{4}$$

After receiving a bad signal, the posterior is

$$\mu_B = \text{Prob}(X_H \mid S_B) = \frac{[\theta(1 - p) + (1 - \theta)(1 - z)]\alpha}{1 - z}. \tag{5}$$

The investment problem is interesting because the paper assumes

$$\mu_G X_H + (1 - \mu_G)X_L > 0 > \mu_B X_H + (1 - \mu_B)X_L. \tag{6}$$

Interpretation: A good signal makes investment efficient; a bad signal makes not investing efficient. Therefore, if managers simply maximized firm value and signals were used efficiently, actions would reveal signals.

3.5 Signal correlation among smart managers

The key modeling assumption is that smart managers' errors are correlated. If both managers are smart, they receive the same signal. If one is smart and one is dumb, or if both are dumb, the dumb signal component is independent.

Interpretation: This is the core force behind reputational herding. If smart managers tend to receive the same signal, then disagreeing with another manager makes the labor market suspect that one is not smart. Correlated prediction errors generate a "sharing-the-blame" motive.

3.6 Labor-market evaluation

The labor market updates the posterior probability that a manager is smart after observing:

- the manager's action;
- the earlier manager's action;
- the realized state X_H or X_L.

Let $\widehat{\theta}(\cdot)$ denote the labor-market posterior that the manager is smart.

Interpretation: The manager's utility is tied to reputation. Even if the project has real economic payoffs, the key distortion arises because the manager wants to maximize perceived ability, not only project value.

3.7 Notation extracted from the paper

LIST OF NOTATION

Symbol	Meaning
x_H	Investment proceeds in "high" state
x_L	Investment proceeds in "low" state
α	*Ex ante* probability of high state
s_G	Good signal
s_B	Bad signal
p	Prob($s_G \mid x_H$, smart)
q	Prob($s_G \mid x_L$, smart)
z	Prob($s_G \mid x_H$, dumb) = Prob($s_G \mid x_L$, dumb)
θ	*Ex ante* probability that manager is smart
$\hat{\theta}$	Posterior probability that manager is smart
μ_G	Prob($x_H \mid s_G$)
μ_B	Prob($x_H \mid s_B$)
Π	Profit passed up in herding equilibrium
R	Reputational benefit to herding
L	Wage that can be earned in outside opportunity

Original paper's list of notation.

Read:

- The notation table collects the model's state payoffs, signal probabilities, posterior beliefs, and reputational variables.
- X_H and X_L are real investment payoffs; S_G and S_B are private signals.
- p and q describe the informativeness of smart managers' signals.
- z is the signal probability for dumb managers and is chosen so that the signal's unconditional distribution does not reveal type.
- Π is the real profit passed up because of herding, while R is the reputational benefit from herding.

Detailed interpretation: The notation shows the paper's two-layer structure. One layer is the real investment problem, summarized by X_H, X_L, and the posterior probabilities μ_G and μ_B. The other layer is the career-concern problem, summarized by $\widehat{\theta}$, R, and L. Herding occurs because the second layer distorts decisions in the first.

Economic message: The table makes clear that the model is not about irrational mistakes. Managers know the signal structure and understand Bayesian updating. The inefficiency arises because their payoff from reputation can dominate the payoff from using their private signal.

4 Models and theoretical specifications

4.1 Efficient benchmark without reputational distortion

If managers cared only about firm profits, each manager would invest if the expected payoff conditional on available information is positive. Ignoring the action of other managers, the signal-contingent rule is:

$$\text{Invest after } S_G, \qquad \text{do not invest after } S_B. \tag{7}$$

Interpretation: This benchmark uses the information in private signals efficiently. Herding is measured against this benchmark: when B receives S_G but does not invest because A did not invest, useful information is lost.

4.2 Reputational payoff and labor-market updating

Manager B's reputation depends on the labor market's posterior belief that B is smart. In a candidate separating equilibrium, actions map one-to-one into signals. Hence the labor market interprets B's action as a report of the signal.

For example, if A is believed to have observed S_B, B is believed to have observed S_G, and the high state occurs, the posterior is written

$$\widehat{\theta}(S_B, S_G, X_H). \tag{8}$$

Interpretation: If B contradicts A and is right, that can help reputation. But if smart managers' signals are correlated, contradicting A is inherently suspicious: it suggests B may be dumb or may have received an idiosyncratic noisy signal.

4.3 Why correlated errors create the payoff to imitation

Under correlated smart signals, the event (S_B, S_G) is impossible if both managers are smart. Therefore, when B's inferred signal differs from A's inferred signal, the labor market assigns lower probability to B being smart.

The paper's Bayesian updating implies two important inequalities:

$$\widehat{\theta}(S_G, S_G, X_H) > \widehat{\theta}(S_B, S_G, X_H), \tag{9}$$

$$\widehat{\theta}(S_G, S_G, X_L) > \widehat{\theta}(S_B, S_G, X_L). \tag{10}$$

Interpretation: Holding the realized state fixed, B looks better when he agrees with A. This is the reputational payoff to conformity. It exists even when B's private information says that disagreement would be economically correct.

4.4 Signal-contingent equilibrium cannot exist: Proposition 1

Proposition 1. There does not exist a continuation equilibrium in which B's investment decision depends on B's signal. The only possible continuation equilibria are pooling equilibria: B mimics A regardless of B's signal, or B contradicts A regardless of B's signal.

Detailed interpretation: The proof starts by assuming a separating equilibrium in which B invests after S_G and does not invest after S_B. It then computes the labor market's Bayesian posteriors. Because disagreement with A is interpreted as evidence against being smart, a B who receives S_G after A did not invest prefers to mimic A rather than reveal disagreement. Thus the candidate separating equilibrium unravels.

Economic message: The result is powerful because it says that reputation can shut down private information aggregation entirely. The problem is not that B cannot interpret his signal. The problem is that B's signal revelation changes how the labor market evaluates him.

4.5 Herding equilibrium with reasonable beliefs: Proposition 2

Proposition 2. There exists a continuation equilibrium in which B always mimics A: B invests if and only if A invests. The equilibrium is supported by reasonable off-equilibrium beliefs: if B deviates by investing after A did not, the labor market believes he observed S_G; if B deviates by not investing after A invested, the labor market believes he observed S_B.

Detailed interpretation: In the herding equilibrium, B's action conveys no information about B's signal. The labor market cannot infer whether B observed S_G or S_B from B's action because both types of signal recipients choose the same action. Deviating is unattractive because the labor market's posterior under deviation is not high enough to compensate for the reputational safety of following A.

Economic message: Herding is self-enforcing. Once the labor market expects B to follow A, B's action cannot credibly reveal his signal. The observed consensus then becomes uninformative, even though each manager individually has information.

4.6 Overall equilibrium: Proposition 3

Proposition 3. There exists an equilibrium of the overall game in which manager A invests if and only if A receives S_G, while manager B always mimics A regardless of B's signal.

Detailed interpretation: Manager A moves first and cannot compare himself to an earlier manager, so A's action can be signal-contingent. Manager B, however, faces a reputational comparison with A. B's private signal is therefore suppressed. The equilibrium has one informative action followed by a herd.

Economic message: The first mover may reveal information, but later movers may stop revealing theirs. Aggregate decisions can therefore become less informative precisely when more decision makers are involved.

4.7 Profit motives and the reputational benefit to herding

The paper defines Π as the real profit passed up in a herding equilibrium and R as the reputational gain from herding. Managers who care only about reputation herd whenever $R > 0$. If they also care about profits, efficient behavior can be restored when the real profit loss Π is large enough relative to R.

$$\text{Herding if reputational benefit } R \text{ exceeds weighted profit loss } \Pi. \tag{11}$$

Interpretation: Profit-based compensation does not eliminate herding in all states, but it shrinks the parameter region in which herding is optimal. The most socially costly herding episodes are less likely if managers have sufficiently strong short-term profit incentives.

4.8 Limited liability

Limited liability changes reputational incentives because a manager who makes an unconventional bad decision may be punished only down to a floor wage or outside option L. If wages cannot fall below L, then downside reputational risk is truncated.

Interpretation: Limited liability can reduce the cost of being wrong alone. This may weaken herding, although the effect depends on where the manager is in his career and how far current wages are above the outside option.

4.9 Relative ability concerns

The baseline model assumes managers care about absolute reputation: the posterior probability of being smart. If managers instead care about relative rank, the incentive to mimic can change.

Interpretation: When relative ranking matters, a manager may prefer to differentiate himself from peers rather than blend into the herd. This can reduce the reputational payoff to imitation.

4.10 Alternative definitions of talent

The model defines talent as the ability to receive an informative signal. If talent were instead defined as the ability to interpret and act on independent information, then agreement with others might not be as favorable for reputation.

Interpretation: Herding depends on the labor market's theory of what smart managers look like. If smart managers are expected to discover similar insights, conformity is rewarded. If smart managers are expected to generate distinct insights, conformity may be penalized.

5 Identification and theoretical design logic

5.1 What the model isolates

- The model isolates career-concern incentives from irrational imitation.
- Managers know the signal structure and update rationally.
- The labor market also updates rationally.
- The inefficient outcome arises because reputational payoffs are not aligned with social information aggregation.

Interpretation: The paper's identification logic is theoretical: hold beliefs rational and show that inefficient imitation can still occur. This makes herding a robust agency phenomenon rather than a psychological anomaly.

5.2 Why sequential decisions matter

- B observes A's action before acting.
- A's action becomes a public benchmark for the labor market's evaluation of B.

- B's private information is evaluated not only by whether it is correct but also by whether it agrees with A's inferred signal.
- The sequential structure generates reputational externalities between managers.

Interpretation: Simultaneous reporting could reduce the problem because managers would not have a public prior action to imitate. This is why committee voting order matters in the paper's organizational application.

5.3 Why correlated smart-manager errors matter

- If smart managers' signals were independent, disagreement would not necessarily imply low ability.
- If smart managers' signals are correlated, disagreement with another apparently informed manager is bad news about one's type.
- This creates the labor-market reward for conventionality.

Interpretation: The model does not require perfect signal correlation. Some correlation in smart managers' prediction errors is enough to make imitation valuable.

5.4 What the paper cannot fully prove

- The paper is theoretical, so it does not estimate the size of herding costs empirically.
- The baseline model assumes fixed investment prices and does not solve a full asset-market clearing model.
- The model assumes a stylized two-type signal structure.
- The equilibrium refinement dismisses the perverse contradiction equilibrium as unreasonable, but equilibrium selection remains conceptually important.
- The paper does not model all organizational mechanisms that may mitigate herding, such as formal scoring rules, team compensation, anonymous voting, or explicit accountability systems.

Interpretation: These limitations do not weaken the core contribution: the paper proves that rational reputation concerns can make informative managers ignore their own signals.

6 Section-by-section study guide

6.1 Introduction

The introduction motivates the problem using Keynes's discussion of professional investors and the idea that it is safer to fail conventionally than to succeed unconventionally. The authors connect this idea to bank lending, corporate investment, money management, and organizational decision making.

How to read it: Focus on the distinction between irrational group psychology and rational career concerns. The whole paper formalizes the latter.

6.2 Section I: The model

Section I sets up two sequential managers, smart/dumb types, good/bad signals, and correlated prediction errors. It defines the posterior probabilities and assumes that a good signal makes investment privately efficient while a bad signal makes noninvestment privately efficient.

How to read it: The most important assumption is not the binary signal structure; it is the correlation of smart managers' signals.

6.3 Section II: Herd behavior

Section II proves the main propositions. Separating behavior by B cannot be sustained because B's reputation is better when he agrees with A. A herding equilibrium exists. An overall equilibrium exists in which A acts on his signal and B follows A.

How to read it: Proposition 1 is the key impossibility result. Proposition 2 provides the natural herding equilibrium. Proposition 3 embeds the continuation equilibrium in the full game.

6.4 Section III: Countervailing forces

Section III explains how profit motives, limited liability, relative ability concerns, and alternative definitions of talent can reduce or modify herding incentives.

How to read it: This section clarifies that the model is not a claim that herding always occurs. Herding is most powerful when reputation dominates short-term profits and ability is judged by conventional correctness.

6.5 Section IV: Applications

Section IV applies the model to corporate investment, stock markets, and decisions within firms. It emphasizes bank lending to LDCs, investment managers' stock choices, and internal committee decisions.

How to read it: This section turns the model into empirical and organizational hypotheses: look for imitation in settings where individual performance is hard to evaluate and mistakes are common across peers.

6.6 Section V: Conclusion

The conclusion summarizes the mechanism and emphasizes that herd behavior can arise in many contexts. The authors also discuss how organizational design can either amplify or reduce herding.

How to read it: The conclusion is especially important for the paper's organizational implications: bottom-up information flow and simultaneous reporting can preserve private information.

7 Tables, figures, and theoretical result blocks

7.1 Original List of Notation

LIST OF NOTATION

Symbol	Meaning
x_H	Investment proceeds in "high" state
x_L	Investment proceeds in "low" state
α	*Ex ante* probability of high state
s_G	Good signal
s_B	Bad signal
p	Prob($s_G \mid x_H$, smart)
q	Prob($s_G \mid x_L$, smart)
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θ	*Ex ante* probability that manager is smart
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μ_G	Prob($x_H \mid s_G$)
μ_B	Prob($x_H \mid s_B$)
Π	Profit passed up in herding equilibrium
R	Reputational benefit to herding
L	Wage that can be earned in outside opportunity

List of notation from Scharfstein and Stein (1990).

Read:

- The table separates real payoff variables (X_H, X_L) from informational variables (S_G, S_B, p, q, z).
- It also defines managerial ability beliefs (θ, $\widehat{\theta}$), signal posteriors (μ_G, μ_B), and reputational/profit variables (R, Π, L).
- The table helps track the key comparison: real project payoffs versus reputational payoffs.

Detailed interpretation: The notation table is not a results table, but it is the closest object in the paper to a compact summary of the model. It reveals how the authors organize the problem into three linked objects: project fundamentals, information quality, and career concerns. The herding mechanism operates exactly because these three objects are not aligned.

Economic message: The manager is not maximizing a single objective. He cares about how the labor market interprets his action. The variables R and Π summarize the central tradeoff: protect reputation or use private information to raise firm value.

7.2 Result Block 1: Proposition 1 – no signal-contingent continuation equilibrium

Read:

- Proposition 1 states that B cannot have a continuation equilibrium in which B's investment depends on B's own signal.
- If B contradicts A, the labor market infers that B's signal differs from A's signal.
- Since smart signals are correlated, differing from A is negative information about B's ability.

- The reputational penalty from disagreement makes B prefer to mimic A.

Detailed interpretation: This proposition is the paper's most important formal result. It does not merely say that B may sometimes imitate A. It says that, under the model's assumptions, signal-contingent behavior by B cannot be sustained in equilibrium. Private information is therefore not just noisy; it is strategically suppressed.

Economic message: Information aggregation fails because agreement is reputationally valuable. A manager may ignore a positive signal after observing that an earlier manager did not invest, not because he thinks the project is bad, but because standing apart from the earlier manager is bad for his career.

7.3 Result Block 2: Proposition 2 – existence of herding equilibrium

Read:

- Proposition 2 shows that there is a continuation equilibrium in which B mimics A regardless of B's signal.
- If A invests, B invests; if A does not invest, B does not invest.
- B's action conveys no information about B's private signal.
- The equilibrium is supported by reasonable off-equilibrium beliefs about deviations.

Detailed interpretation: The proof clarifies why the herd is self-sustaining. Once B is expected to pool, B's action no longer reveals information. A deviation is interpreted through off-equilibrium beliefs, and the reputational gain from deviation is too small relative to the safety of following A.

Economic message: The visible consensus in the market or organization can be misleading. It may look as if many managers independently agree, when in fact only the first action is informative and all later actions are reputational imitation.

7.4 Result Block 3: Proposition 3 – overall equilibrium

Read:

- Proposition 3 embeds the herding continuation equilibrium into the full game.
- A acts efficiently on A's signal.
- B mimics A regardless of B's signal.
- The result extends to more than two managers: later managers may also learn nothing from intermediate herd actions.

Detailed interpretation: The first mover's action is informative because the first mover has no predecessor to imitate. Later actions are potentially uninformative. The model therefore generates cascades: information revealed early can dominate, and later private information can be wasted.

Economic message: Sequential decision making can be fragile. Once the herd starts, adding more decision makers does not necessarily add more information. More voices can produce less learning if each voice is trying to protect reputation.

7.5 Result Block 4: Countervailing forces

Read:

- Profit-based incentives reduce herding when the real profit loss from herding is large.
- Limited liability may reduce the downside reputational cost of being wrong alone.
- Relative performance concerns can make differentiation more attractive.
- A different labor-market definition of talent can change whether imitation is rewarded.

Detailed interpretation: This block is important because it makes the model empirically nuanced. Herding is strongest in settings where managers care heavily about reputation, where their compensation is weakly tied to short-term profits, where mistakes are hard to attribute to individuals, and where smart people are expected to agree.

Economic message: Institutional design matters. Compensation, voting rules, peer comparison, and the definition of managerial ability can change whether private information is used or suppressed.

7.6 Result Block 5: Applications

Read:

- In corporate investment, managers may follow peers into similar projects or lending strategies to avoid blame for being unconventional.
- In stock markets, professional money managers may buy what others are buying and sell what others are selling, amplifying price movements.
- Within firms, committees may waste information if early speakers influence later speakers.
- The paper suggests bottom-up information flow as a partial remedy.

Detailed interpretation: The applications show that the model is not just about investment in physical capital. It applies whenever agents are evaluated by outsiders who cannot fully observe private information and who use conformity as a signal of competence.

Economic message: The model has broad implications for financial crises, investment booms, analyst behavior, mutual fund trading, board decisions, and corporate committees. Herding is a governance problem, not just a market anomaly.

8 Detailed result map

- **Model setup:** sequential managers; high/low states; good/bad signals; smart/dumb types; correlated smart signals.
- **Efficient benchmark:** invest after S_G, do not invest after S_B.
- **Bayesian updating:** the labor market updates ability using both outcomes and agreement/disagreement with other managers.
- **Key distortion:** agreeing with an earlier manager is reputationally valuable when smart managers' signals are correlated.
- **Proposition 1:** B cannot have a signal-contingent continuation equilibrium.

- **Proposition 2:** herding continuation equilibrium exists.
- **Proposition 3:** overall equilibrium exists with A signal-contingent and B herding.
- **Countervailing forces:** profit motives, limited liability, relative ranking, and alternative ability definitions can weaken herding.
- **Applications:** corporate investment, bank lending, money management, and internal firm decision making.

9 Short summary

- The paper formalizes rational herd behavior under career concerns.
- Managers may ignore their own private signals to protect reputation.
- Smart managers receive informative signals; dumb managers receive uninformative signals.
- Smart managers' prediction errors are correlated, so agreement with other managers is favorable for perceived ability.
- Manager A moves first and can reveal information through action.
- Manager B observes A and receives his own signal.
- B's action cannot be signal-contingent in equilibrium because disagreement with A is reputationally costly.
- A herding equilibrium exists in which B always mimics A.
- The herd destroys information aggregation: B's private signal is wasted.
- The model explains why managers may prefer conventional mistakes to unconventional success.
- Profit incentives and organizational design can reduce, but not necessarily eliminate, herding.
- Applications include corporate investment, bank lending, stock markets, and committee decision making.

10 Conclusion

10.1 Core conclusion

Scharfstein and Stein show that herd behavior can be a rational response to career concerns. Managers who are evaluated by a labor market may prefer to imitate others rather than act on private information, because being wrong alone is worse for reputation than being wrong with the crowd. In equilibrium, private information can be suppressed and aggregate investment can be inefficient.

10.2 Economic intuition

The key intuition is that agreement is informative about ability when skilled managers are expected to receive similar signals. If manager B disagrees with manager A, the labor market may infer that B is not smart. This reputational cost can dominate the real economic value of B's private signal. Thus the manager does not simply ask, "Is this investment profitable?" He also asks, "What will this action make the market think about me?"

10.3 Incremental contribution to economics and finance

The paper's enduring contribution is to transform herd behavior from an informal psychological phenomenon into an equilibrium outcome of rational career concerns. It links the economics of information, labor-market reputation, and corporate investment. It also helps explain why dispersed information may fail to aggregate in markets and organizations, even when agents are rational and information is available.

10.4 Implications for corporate governance

The model suggests that boards and firms should be careful when evaluating managers by relative conventionality. If managers are rewarded for conforming to industry norms, they may ignore valuable private information. Governance systems should therefore reward well-supported dissent and not only ex post outcomes. This is especially relevant for long-horizon investments where outcomes are noisy and attribution is difficult.

10.5 Implications for financial markets

The paper provides a rational foundation for trend following and correlated institutional behavior. Professional money managers may buy assets that others are buying, not because they lack information, but because deviating from the consensus is reputationally costly. This can amplify booms, busts, and price movements unrelated to fundamentals.

10.6 Implications for organizational design

The paper implies that information aggregation depends on procedure. Sequential meetings can create cascades; simultaneous reporting can reduce them. Bottom-up information flow may be superior to top-down flow because junior or less secure managers can reveal information before senior reputations shape the discussion.

10.7 Limitations

The model is stylized. It uses binary signals, binary states, and a sequential two-manager structure. It does not estimate the magnitude of herd behavior empirically. It abstracts from asset-price feedback and from many real organizational mechanisms. Nevertheless, its logic is robust: when reputational evaluation rewards conventionality, information aggregation can fail.

10.8 Takeaway

The central lesson is that the problem of herding is not only that people imitate irrationally. It is that rational agents may choose imitation when reputation is at stake. Efficient decision making requires institutions that make it safe to use private information, especially when that information contradicts the crowd.